

ANDILE CLIFFORD MATHE

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PROFESSIONAL SUMMARY

Software developer and AI engineer with a background in Information Technology and hands-on experience delivering production-grade systems across the full software development lifecycle — from requirements and design through to testing, deployment, and support. Comfortable moving between backend APIs, frontend interfaces, workflow automation, and agentic AI pipelines, with a focus on components that are robust, maintainable, and scalable. Currently building LangChain/LangGraph-based AI agents at an internship, and independently architecting and shipping a live food delivery platform end-to-end — covering real-time dispatch automation, role-based dashboards, payment integration, and cloud deployment. Strong foundation in clean code, RESTful API design, database management, troubleshooting, and collaborative, Agile development using Git.

TECHNICAL SKILLS

Languages: C# (.NET Core), Java, JavaScript (Node.js, Express.js), Python, PHP, Dart, HTML/CSS

Frontend: React, Vite, Angular, Tailwind CSS

AI & Automation: LangChain, LangGraph (ReAct agents), RAG pipelines, ChromaDB, HuggingFace embeddings, LLM integration (Ollama, Groq, Claude API), LangSmith, n8n workflow orchestration, webhook-driven event systems

Backend & APIs: FastAPI, Flask, RESTful API design, third-party API integration (Twilio, PayFast, Airtable, Clerk)

Databases: PostgreSQL, MySQL, MariaDB, Firebase, Airtable

DevOps & Infrastructure: Docker, Linux (Ubuntu), Nginx, CI/CD pipelines, VPS deployment, SSL/TLS (Let's Encrypt), Git/GitHub

Quality Assurance: Unit testing (Vitest, k6 load testing), debugging, performance optimisation

Methodologies & Practices: Agile/Scrum, Software Development Life Cycle (SDLC), requirements gathering (functional & non-functional), system design and architecture, technical documentation, user acceptance testing (UAT), stakeholder communication, code review, REST API design principles

PROFESSIONAL EXPERIENCE

AI Engineer Intern (Agentic Systems) — New Island Technologies, Midrand 2026 — Present

- Designing and building an agentic AI document-drafting system using LangChain and LangGraph (ReAct architecture), integrated with a FastAPI backend and React/Vite frontend.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and HuggingFace embeddings (all-MiniLM-L6-v2) to ground AI output strictly in source documents and eliminate hallucination.
- Integrated Groq LLM inference (llama-3.3-70b-versatile) with LangSmith tracing and evaluation for monitoring agent consistency and output quality.
- Built a human-in-the-loop approval workflow ensuring no automated action is taken without explicit user review.
- Explored and documented BMC Helix Innovation Studio for enterprise ITSM/ITOM workflow automation, mapping client WhatsApp communication flows to structured ticket creation pipelines.

Software Developer (Practical Placement) — FNETT, Polokwane 2025 — One Year

- Designed, coded, and maintained backend services in a live production environment, delivering working software to defined requirements and coding standards.

- Led the full backend redesign and migration from PHP to Node.js and Express.js, improving maintainability, re-usability, and performance.
- Built and maintained RESTful APIs integrated with a Flutter mobile application and multiple third-party services, with a focus on reliability and fault tolerance.
- Managed MariaDB databases including schema design, query optimisation, and data integrity maintenance.
- Containerised services using Docker and wrote shell scripts to automate monitoring and operational tasks.
- Collaborated with frontend teams (React, Angular) in an Agile environment, contributing to integrated feature delivery and providing guidance to peers during code reviews.

INDEPENDENT PROJECTS

Food Delivery Platform — Full-Stack & Automation — Personal Project (Production) 2026

- Architected and built a complete food delivery platform from scratch — currently in active operation — covering dispatch automation, real-time driver assignment, role-based dashboards, and payment integration.
- Built a real-time dispatch engine in n8n orchestrating 6+ automated workflows: driver proximity matching (Haversine algorithm), offer timeout management, retry escalation (30/60/90-min tiers), and multi-phase WhatsApp notification pipelines via Twilio.
- Designed and implemented role-based React dashboards (Admin, Vendor, Driver) with Clerk authentication, Airtable/PostgreSQL data layers, and real-time order state management.
- Integrated PayFast payment processing with dynamic link generation per order and ITN webhook handling for payment confirmation before dispatch.
- Deployed the full stack to a self-hosted VPS (Ubuntu, Nginx reverse proxy, SSL via Let's Encrypt), with the React frontend on Vercel.
- Built automated GPS driver location tracking via the browser Geolocation API with 30-second Airtable updates, with a planned migration to Traccar for native background tracking.
- Wrote 15+ passing Vitest unit tests and conducted k6 load testing, which identified Airtable rate limiting and led to a vendor data caching solution.
- Used Claude API for agentic task assistance throughout the build, and integrated an n8n MCP server for natural-language workflow management.

EDUCATION

Information Technology

Tshwane University of Technology | 2021 – 2025

- Graduated with multiple subject distinctions.
- Member of the Golden Key International Honour Society — top 15% of the ICT Faculty.

National Senior Certificate (Matric)

Thornview Prestige College | 2019

LANGUAGES

Fluent: English, IsiZulu

Good working knowledge: Afrikaans, Setswana, Sepedi, Xitsonga

REFERENCES

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Additional references available on request.